### Use De Morgan's laws to write negations for the statements in 25–30.

**45.** Determine whether the statements in (a) and (b) are logically equivalent.

**a.** Bob is both a math and computer science major and Ann is a math major, but Ann is not both a math and computer science major.

**b.** It is not the case that both Bob and Ann are both math and computer science majors, but it is the case that Ann is a math major and Bob is both a math and computer science major.

First, let's break down the statement forms into individual statement variables.

p = Bob is a math major
q = Bob is a computer science major

r = Ann is a math major
s = Ann is a computer science major

Now we can create statement forms:

**a.** $p \wedge q \wedge r \wedge \sim(r \wedge s) \equiv p \wedge q \wedge r \wedge (\sim r \vee \sim s)$
**b.** $\sim(p \wedge r \wedge q \wedge s) \wedge r \wedge p \wedge q \equiv (\sim p \vee \sim r \vee \sim q \vee \sim s) \wedge r \wedge p \wedge q$

In order to determine whether the statements are logically equivalent, one method is to draft a truth table for each statement form.

**a.** p ∧ q ∧ r ∧ (~r ∨ ~s)

| **~r** | **~s** | **(~r ∨ ~s)** | **p** | **q** | **r** | **p ∧ q ∧ r ∧ (~r ∨ ~s)** |
|---|---|---|---|---|---|---|
| F | F | F | T | T | T | F |
| F | T | T | T | T | T | T |
| T | F | T | T | T | F | F |
| T | T | T | T | T | F | F |
| F | F | F | T | F | T | F |
| F | T | T | T | F | T | F |
| T | F | T | T | F | F | F |
| T | T | T | T | F | F | F |
| F | F | F | F | T | T | F |
| F | T | T | F | T | T | F |
| T | F | T | F | T | F | F |
| T | T | T | F | T | F | F |
| F | F | F | F | F | T | F |
| F | T | T | F | F | T | F |
| T | F | T | F | F | F | F |
| T | T | T | F | F | F | F |

**b.** (~p ∨ ~r ∨ ~q ∨ ~s) ∧ r ∧ p ∧ q

| **~p** | **~r** | **~q** | **~s** | **r** | **p** | **q** | **(~p ∨ ~r ∨ ~q ∨ ~s) ∧ r ∧ p ∧ q** |
|---|---|---|---|---|---|---|---|
| F | F | F | F | T | T | T | F |
| F | F | F | T | T | T | T | T |
| F | T | F | F | F | T | T | F |
| F | T | F | T | F | T | T | F |
| F | F | T | F | T | T | F | F |
| F | F | T | T | T | T | F | F |
| F | T | T | F | F | T | F | F |
| F | T | T | T | F | T | F | F |
| T | F | F | F | T | F | T | F |
| T | F | F | T | T | F | T | F |
| T | T | F | F | F | F | T | F |
| T | T | F | T | F | F | T | F |
| T | F | T | F | T | F | F | F |
| T | F | T | T | T | F | F | F |
| T | T | T | F | F | F | F | F |
| T | T | T | T | F | F | F | F |

Now we can compare the two statements (a) and (b):

| $p \wedge q \wedge r \wedge (\sim r \vee \sim s)$ | $(\sim p \vee \sim r \vee \sim q \vee \sim s) \wedge r \wedge p \wedge q$ |
|---|---|
| F | F |
| T | T |
| F | F |
| F | F |
| F | F |
| F | F |
| F | F |
| F | F |
| F | F |
| F | F |
| F | F |
| F | F |
| F | F |
| F | F |
| F | F |
| F | F |

When examining the statement forms we can now conclude that the statement forms are logically equivalent.

**Solution:** Yes, the statement forms are logically equivalent, because they each share identical truth values for every possible combination.